THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · EASY LEVEL UNIT TEST SET 3

Chapters 5–7: **Arithmetic Progressions, Triangles, Coordinate Geometry** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. In an AP with $a = 8$ and $d = -2$, the 2nd term is: 1

(a) 6 **(b)** 10 **(c)** -16 **(d)** 4

2. The diagonal of a rectangle with sides 5 and 12 is: 1

(a) 169 **(b)** 13 **(c)** 17 **(d)** 7

3. The distance between the points $(5, 0)$ and $(0, 12)$ is: 1

(a) 7 **(b)** 17 **(c)** 13 **(d)** 169

4. The 1st term of an AP is 7 and the 3rd term is 17. The common difference is: 1

(a) 5 **(b)** $\frac{5}{2}$ **(c)** 12 **(d)** 10

5. A triangle has sides 3, 4, 5. It is right-angled if: 1

(a) $3 + 4 = 5$ **(b)** $3^2 + 4^2 = 5^2$ **(c)** $3^2 = 4^2 + 5^2$ **(d)** $3 \times 4 = 5$

6. The abscissa of the point $(7, -3)$ is: 1

(a) 7 **(b)** -7 **(c)** -3 **(d)** 4

7. The common difference of the AP $7, 12, 17, 22, \ldots$ is: 1

(a) -5 **(b)** 10 **(c)** 5 **(d)** 6

8. In a right triangle with legs 8 and 15, the hypotenuse is: 1

(a) 7 **(b)** 17 **(c)** 23 **(d)** 289

9. The distance of the point $(8, 15)$ from the origin is: 1

(a) 289 **(b)** 23 **(c)** 7 **(d)** 17

10. The 6th term of the AP $2, 9, 16, 23, \ldots$ is: 1

(a) 37 **(b)** 30 **(c)** 44 **(d)** 42

11. The ratio of corresponding sides of two similar triangles is 3:4. Their perimeters are in the ratio 3: 1

(a) 3 **(b)** 4 **(c)** 12 **(d)** 7

12. The midpoint of the join of $(-2, 3)$ and $(4, 7)$ is: 1

(a) $(2, 10)$ **(b)** $(1, 6)$ **(c)** $(2, 5)$ **(d)** $(1, 5)$

13. The sum of the first 6 terms of $5, 10, 15, 20, \ldots$ is: 1

(a) 75 **(b)** 140 **(c)** 210 **(d)** 105

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14. In $\triangle ABC$, $DE \parallel BC$, $AD = 1$, $DB = 5$, $AE = 3$. Then $EC =$ 1

(a) 16 **(b)** 3 **(c)** 15 **(d)** 5

15. The point $(-3, -1)$ lies in the: 1

(a) second quadrant **(b)** fourth quadrant **(c)** third quadrant **(d)** first quadrant

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. Find the hypotenuse of a right triangle with legs 9 cm and 12 cm. 2

17. Find the distance between the points $(1, 2)$ and $(4, 6)$. 2

18. Find the common difference of the AP $20, 17, 14, 11, \ldots$ 2

19. The sides of two similar triangles are in the ratio 3:4. Find the ratio of their areas. 2

20. Find the midpoint of the segment joining $(-4, 3)$ and $(6, -1)$. 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. Find the midpoint of the segment joining $(-6, -2)$ and $(4, 8)$. 3

22. Which term of the AP $3, 10, 17, 24, \ldots$ is 59? 3

23. The sides of a triangle are 5 cm, 7 cm and 9 cm. Is it a right triangle? If so, name the right angle. 3

24. Find the coordinates of the point which divides the join of $(2, 1)$ and $(8, 7)$ internally in the ratio $1 : 1$. 3

OR

Find the 8th term of the AP $2, 9, 16, 23, \ldots$ 3

25. In a right triangle the two perpendicular sides are 8 cm and 15 cm. Find the hypotenuse and the perimeter. 3

OR

Find the distance between the points $(-3, -2)$ and $(2, 10)$. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. Find the area of the triangle with vertices $(1, 1)$, $(5, 1)$ and $(3, 6)$. 5

OR

In an AP, $a = 9$, $d = 3$, $n = 14$. Find a_n and S_n. 5

27. A pole 5 m high casts a shadow 3 m long. At the same time a tower casts a shadow 21 m long. Find the height of the tower. 5

OR

Find the centroid of the triangle with vertices $(1, 2)$, $(4, 6)$ and $(7, 4)$. 5

END OF QUESTION PAPER Total: 50 Marks

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